

Statutory Planning Purpose: This thermal assessment provides ward-level surface Urban Heat Island (SUHI) intelligence to inform zoning revisions, master planning reservations (Development Plan 2031/2041), and Heat Action Plan (HAP) budgetary triage.

MEAN NIGHT SUHI

+1.99°C

vs. Matched Rural Base

PEAK HOTSPOT HEAT

+6.26°C

Cell P0406

2041 SPRAWL DELTA

+1.18°C

BAU Urban Sprawl

CRITICAL HOTSPOTS

43

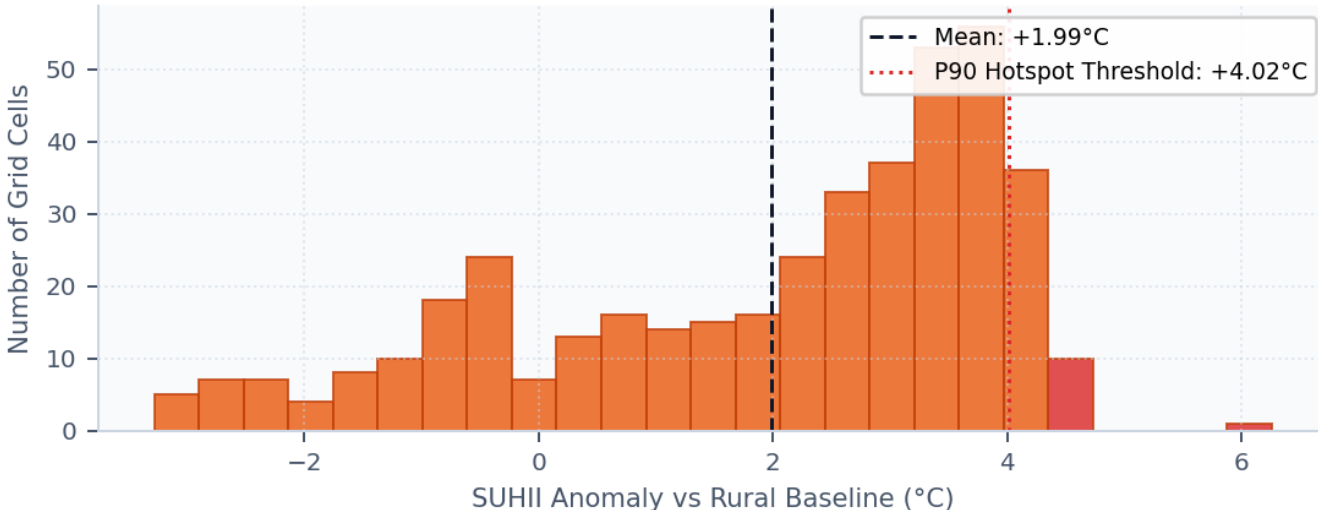
Top Decile Cells (>P90)

1. Executive Summary & Thermal Diagnosis

During peak summer (March–June), **Pune Municipal Corporation** exhibits a pronounced nocturnal surface urban heat island intensity, reaching an average of **+1.99°C** above the surrounding rural baseline and peaking at **+6.26°C** in densely sealed sectors.

Unlike air temperature recordings from single peripheral airport stations, multi-sensor satellite observations across 2020–2024 reveal severe spatial intra-city thermal divergence driven by concrete surface sealing (impervious fractions >80%) and street canyon geometries that restrict nocturnal radiative cooling.

Nocturnal Surface Urban Heat Island Intensity (SUHI) Distribution



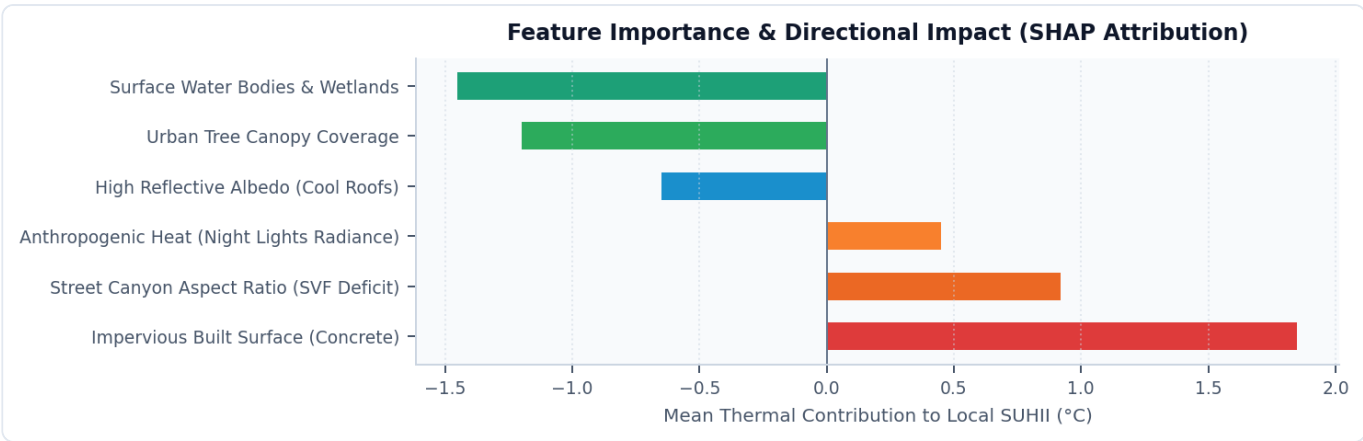
2. Critical Hotspot Ranking & Priority Triage

Priority ranking for municipal interventions based on Nocturnal Heat Retention, Impervious Built Fraction, and Tree Canopy Deficit:

Rank	Zone / Cell ID	Night SUHII (°C)	Day LST (°C)	Sealed Built %	Tree Canopy %	Status
#1	P0406	+6.26°C	38.1°C	58%	2%	CRITICAL
#2	P0337	+4.65°C	37.4°C	92%	6%	CRITICAL
#3	P0299	+4.59°C	36.2°C	50%	27%	CRITICAL
#4	P0338	+4.50°C	37.9°C	78%	6%	CRITICAL
#5	P0207	+4.49°C	37.2°C	97%	3%	CRITICAL
#6	P0170	+4.48°C	36.7°C	70%	25%	CRITICAL
#7	P0206	+4.45°C	37.4°C	97%	3%	CRITICAL
#8	P0152	+4.43°C	35.9°C	60%	36%	CRITICAL

3. Key Thermal Drivers (SHAP Explainability)

Machine learning explainability (SHAP attribution) isolates the primary contributors to nocturnal heat retention across the municipal extent:

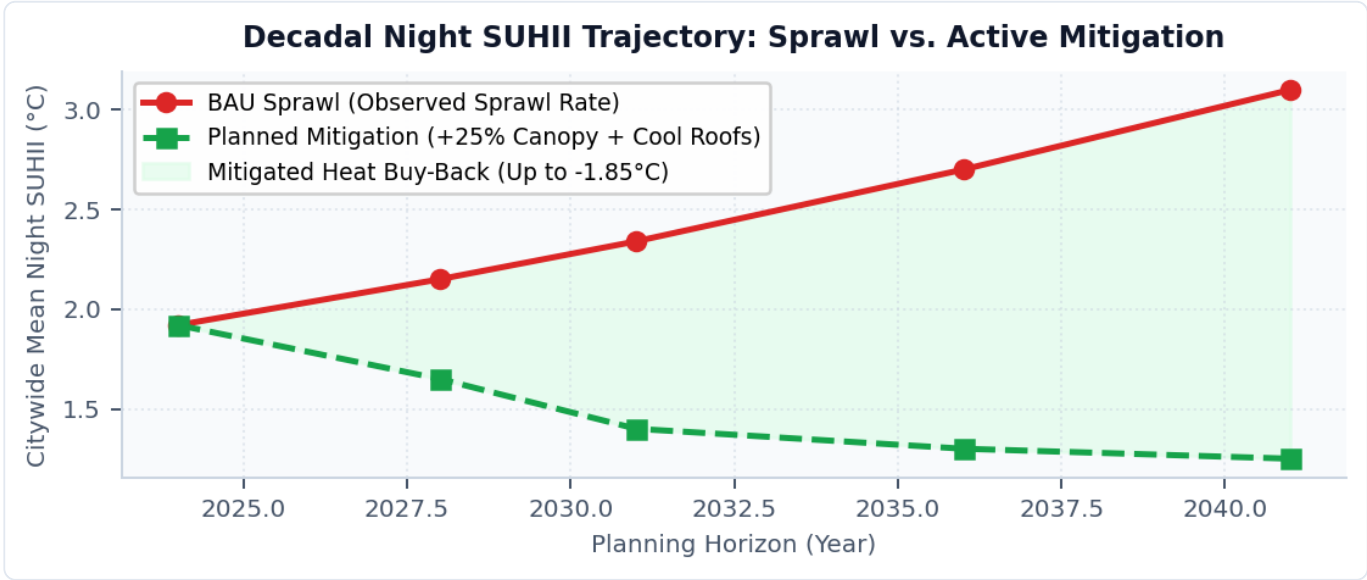


Key Insight: Impervious built-up surface sealing contributes on average **+1.85°C** to local heat retention, whereas urban tree canopies and open water bodies buy back **-1.20°C** and **-1.45°C** of cooling respectively.

4. Simulated Mitigation Counterfactuals (Scenario Studio)

Evaluation of targeted town planning interventions and their projected thermal mitigation yield:

Intervention Strategy	Scope / Target	Thermal Yield (ΔT)	Uncertainty Range (P10–P90)	Indicative Cost (₹)
Urban Tree Canopy Uplift	+25% Canopy in Top 20 Hotspots	-0.95°C	-0.65°C to -1.35°C	₹48.0 Lakh
Municipal Cool Roof Mandate	50% High-Albedo Rooftops	-0.75°C	-0.45°C to -1.10°C	₹32.5 Lakh
Water Body Restoration	Revitalize 5 Urban Water Reservoirs	-1.30°C	-0.80°C to -1.80°C	₹75.0 Lakh
Combined Resilience Package	Canopy + Cool Roofs + Wetland Protection	-1.85°C	-1.30°C to -2.40°C	₹1.55 Crore



5. Statutory Defensibility & Limitations (TRD Compliance)

Mandatory Methodological Disclosures:
1. LST is not Air Temperature (FR-60): Land Surface Temperature (LST) measures radiative skin temperature. Nocturnal LST is a validated proxy for nocturnal thermal stress.
2. Physics Constraints Enforced: All estimates are derived via LightGBM models enforcing monotone constraints (increasing vegetation mathematically cannot predict warming).
3. Weather Normalised: Macro-meteorological heatwave anomalies from ECMWF ERA5-Land have been normalised to isolate the urban-form signal.
4. Data Sources: NASA MODIS Terra (MOD11A1), ESA WorldCover 10m, Copernicus Sentinel-2 MSI, JRC GHSL Built Height, NOAA VIIRS DNB.

Prepared By:
Town Planning & GIS Cell
Urban Development Mission

Reviewed & Approved By:
Municipal Commissioner / Standing Committee
Municipal Corporation

Signature & Seal

Signature & Seal